Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Data Cleaning and Preparation using Power Query

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Data Cleaning and Preparation of Cafe Sales Data using Power Query

GitHub Link:

Problem Statement

A cafe business had accumulated customer transaction data that contained missing values, duplicate Transaction IDs, and inconsistent entries such as invalid placeholder text ("ERROR", "UNKNOWN") and blank fields across key columns, making the raw dataset unreliable for reporting and analysis. Manually inspecting and correcting such issues in a spreadsheet is time-consuming, inconsistent, and error-prone as the volume of transactional data grows. This created the need for a structured, repeatable data-cleaning process using Power Query that could detect data-quality issues, standardize formats, remove duplicates, handle missing and invalid values, and prepare a clean, analysis-ready dataset.

Methodology / Steps Followed

Step 1: Data Collection

- Dataset sourced from Kaggle: "Cafe Sales – Dirty Data for Cleaning Training" (dirty_cafe_sales.csv), available at kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training
- The raw file contains transaction-level records with columns including Transaction ID, Item, Quantity, Price Per Unit, Total Spent, Payment Method, Location, and Transaction Date, across 1,000+ rows
- The dataset is intentionally "dirty," containing missing values, invalid placeholder text ("ERROR", "UNKNOWN"), and inconsistent entries to simulate real-world raw data

Step 2: Data Import

- Imported dirty_cafe_sales.csv into Power BI Desktop using the Get Data → Text/CSV connector
- Verified column headers and row count in the import preview before selecting Transform Data to open the Power Query Editor
- Confirmed that the raw preview already showed "ERROR" and "UNKNOWN" placeholder values and blank cells scattered across multiple columns, matching the description of the dataset on Kaggle

Step 3: Data Profiling

- Enabled Column Quality, Column Distribution, and Column Profile under the View tab in Power Query Editor to assess the health of each column
- The Column Quality report showed Transaction ID and Item as almost fully valid, while Payment Method (recorded initially as "Payment Method Raw") showed a large proportion of empty cells — roughly a third of the column — making it one of the fields most in need of cleaning
- Quantity and Price Per Unit were mostly valid but still contained a small share of empty cells and errors that needed to be addressed
- Identified "ERROR" and "UNKNOWN" as recurring invalid placeholder values embedded across the Item, Quantity, Price Per Unit, Payment Method, Location, and Transaction Date columns

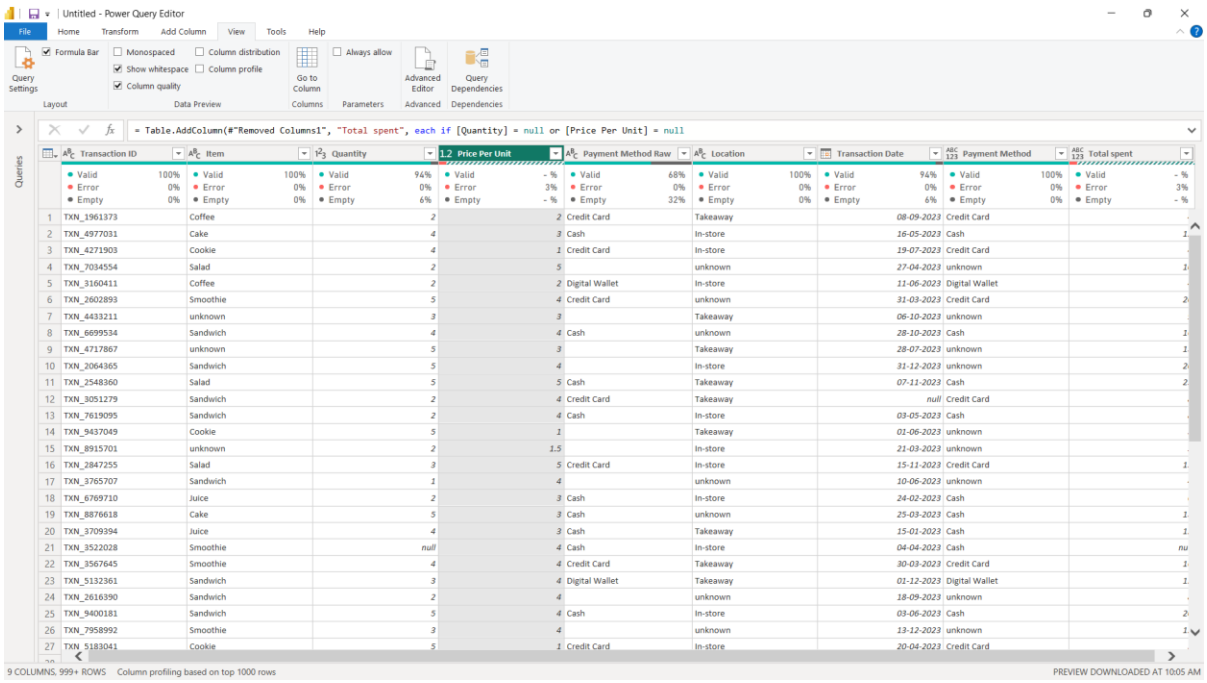
Step 4: Data Cleaning and Transformation

- Replaced invalid placeholder text (“ERROR”, “UNKNOWN”) with null across all affected columns using Replace Values, applied as a series of Replaced Value / Replaced Errors steps in the Applied Steps pane
- Reviewed and handled missing values in Item, Payment Method, and Location so that blanks did not interfere with downstream analysis
- Corrected column data types so that Quantity and Price Per Unit were treated as numeric, Transaction Date was treated as a date, and identifier/text columns such as Transaction ID, Item, Payment Method, and Location were treated as text
- Added a custom Total Spent column that flagged rows where either Quantity or Price Per Unit was null, then recalculated a clean total as Quantity × Price Per Unit to fix rows where the original Total Spent value was an error or missing
- Removed the original erroneous Total Spent column and renamed the corrected column back to “Total Spent” (applied step: Table.RenameColumns(#"Removed Columns2", {{"Total Spent New", "Total Spent"}}))
- Removed duplicate rows based on Transaction ID using Remove Duplicates
- Standardized column naming (e.g., renaming “Payment Method Raw” back to “Payment Method”) and reviewed the Transaction Date column for a consistent format

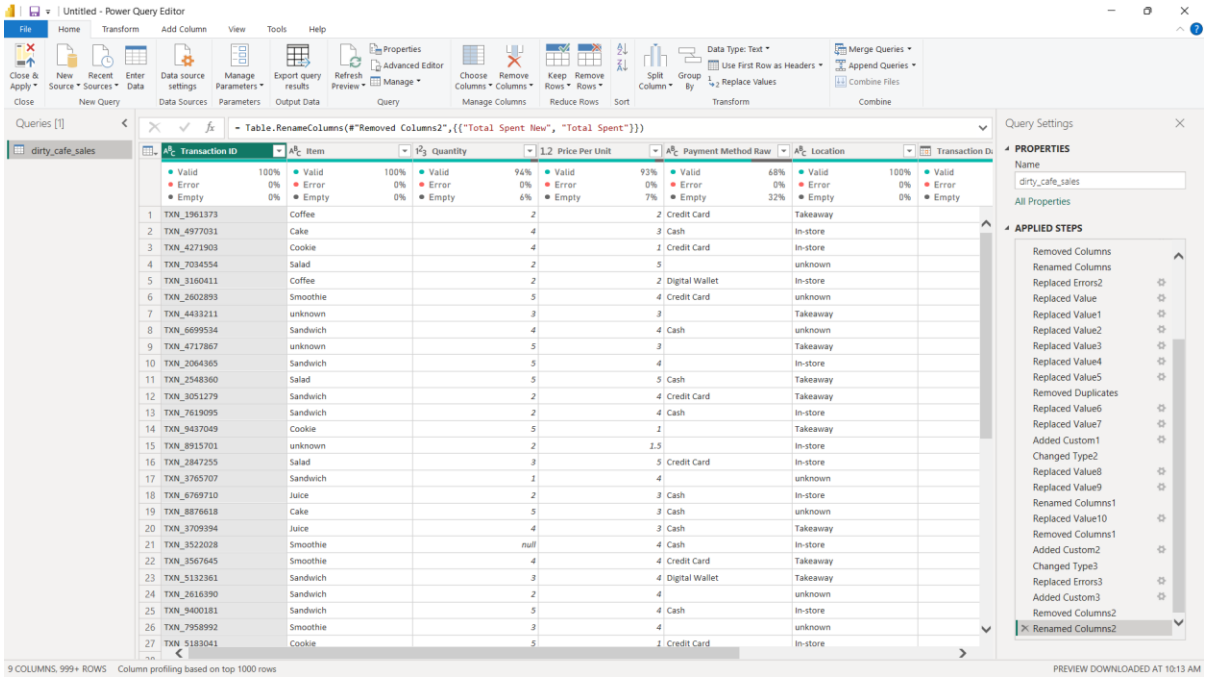
Step 5: Load and Validation

- Re-checked the Column Quality panel after cleaning to confirm a reduced Error/Empty percentage and a higher Valid percentage across the affected columns
- Reviewed the full list of Applied Steps to confirm the cleaning sequence — column removal, renaming, error/value replacement, duplicate removal, type changes, and the custom Total Spent calculation — was captured and reproducible
- Loaded the cleaned dirty_cafe_sales query into the Power BI data model, ready for further analysis and visualization

Output



- Power Query Editor view of the dirty_cafe_sales query with Column Quality, Column Distribution, and Column Profile indicators enabled, showing columns including Transaction ID, Item, Quantity, Price Per Unit, Payment Method Raw, Location, Transaction Date, Payment Method, and Total spent
- Column Quality statistics displayed per column (Valid / Error / Empty percentages); Payment Method Raw stands out with roughly two-thirds valid and about a third empty, the largest gap among the columns shown
- The formula bar shows the Added Custom step used to build the Total spent column, using a conditional expression that checks whether Quantity or Price Per Unit is null before calculating the total



- Applied Steps pane for the dirty_cafe_sales query, listing the full cleaning sequence: Removed Columns, Renamed Columns, multiple Replaced Errors / Replaced Value steps, Removed Duplicates, further Renamed Columns and Removed Columns steps, Added Custom columns, Changed Type steps, and the final Renamed Columns2 step
- The formula bar shows the final rename step, Table.RenameColumns(#"Removed Columns2", {{"Total Spent New", "Total Spent"}}), which restores the recalculated total back to its original column name
- Together, the two views document the before-and-after of the cleaning process: the column-quality diagnosis that identified problem areas, and the reproducible step-by-step transformation that resolved them

Conclusion

This project successfully used Power Query within Power BI Desktop to transform a raw, error-prone cafe sales dataset into a clean, consistent, and analysis-ready table. By profiling column quality, replacing invalid placeholder values such as “ERROR” and “UNKNOWN,” handling missing entries, correcting data types, recalculating the Total Spent field from Quantity and Price Per Unit, and removing duplicate Transaction IDs, the reliability of the dataset was significantly improved. The exercise demonstrated how Power Query's profiling and transformation tools can replace slow, manual, spreadsheet-based cleaning with a repeatable, auditable process suitable for downstream reporting and analysis.

Learning Outcomes

- Learned to use Power Query's Column Quality, Column Distribution, and Column Profile tools to diagnose data-quality issues before cleaning
- Gained practical experience identifying and replacing invalid placeholder values (“ERROR”, “UNKNOWN”) and handling missing/null values
- Understood how to correct column data types and recalculate a derived column (Total Spent) to fix rows with corrupted or missing numeric values

- Practiced removing duplicate records based on a unique identifier (Transaction ID) and standardizing inconsistent column names
- Learned to trace and document transformation logic through Power Query's Applied Steps pane for reproducibility

Business Insights

- A notable share of missing data was concentrated in the Payment Method field, suggesting this field may need better validation at the point of sale to avoid gaps in payment-method analysis
- Placeholder values like “ERROR” and “UNKNOWN” point to upstream data-entry or system-integration issues that should ideally be fixed at the source rather than repeatedly cleaned downstream
- Recalculating Total Spent from Quantity and Price Per Unit exposed rows where the originally recorded revenue figure was unreliable, which could otherwise have understated or overstated sales performance
- Removing duplicate Transaction IDs ensures that revenue and item-count metrics are not artificially inflated by repeated records
- A clean, validated dataset is a prerequisite for any further sales, revenue, or product-performance analysis, reinforcing the value of a dedicated data-cleaning stage before dashboard development